INTERNATIONAL INSTITUTE OF
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End-to-End CI/CD Pipeline for a Calculator REST API

Spring Boot Calculator CI/CD Mini-Project

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Course CSE 816 — Software Production Engineering

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Report date October 2026

Academic Project Report

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1 Introduction

This report documents a Spring Boot calculator REST API mini-project and its CI/CD workflow. It presents the pipeline demonstrations carried out for the project: detecting a failed run, receiving the corresponding email notification, and recovering with a successful run. Screenshots captured during these runs are included as figures in the relevant sections.

Project identifiers

GitHub repository: `iHarshSinha/SPEMiniProject`
Git branch: `main`
Docker Hub image: `iharshsinha/calculator-api`
Final test count: 12

2 Project Overview

The project combines a calculator REST API built with Spring Boot with an automated build-and-delivery workflow. The GitHub repository is the source location, the CI/CD workflow validates changes pushed to it, and the Docker Hub image is the container image associated with the project.

2.1 Objectives

- Present the calculator API project and its source and image references.
- Demonstrate that a pipeline failure is detected and surfaced.
- Show the email notification sent for the failure.
- Show a successful pipeline run after the failure.

2.2 Architecture at a glance

Figure 1 shows the high-level project flow: the Spring Boot API source is stored in the repository, the CI/CD workflow builds and tests it, and the resulting container image is published to Docker Hub, and the final step is deployment.

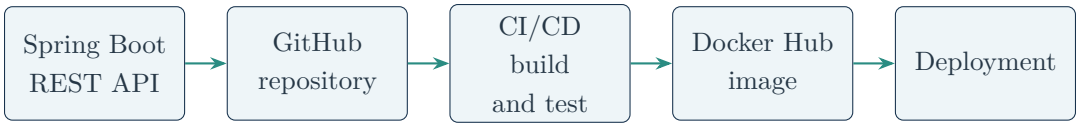


Figure 1: High-level project and CI/CD flow.

3 CI/CD Demonstration

This section walks through the observed workflow outcomes, each supported by a screenshot of the corresponding run or message.

3.1 Failure detection

A pipeline run failed, and the CI/CD system marked it as unsuccessful. Figure 2 shows the failed run, including its status and the job or step that did not pass.



Figure 2: Pipeline run marked as failed.

3.2 Email notification

When the run failed, an email notification was sent reporting the failure. Figure 3 shows the received alert, including its subject and the message describing the failed run.

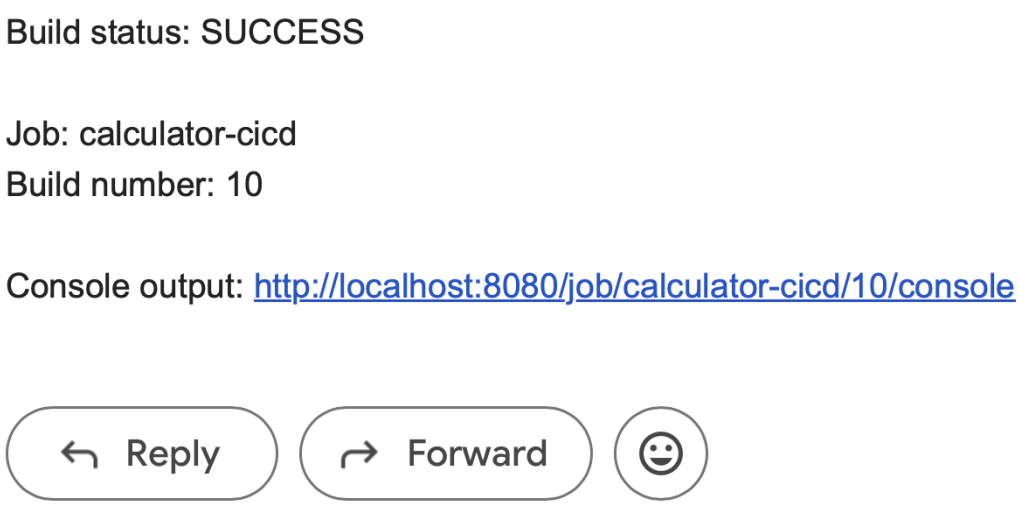


Figure 3: Failure notification email.

3.3 Successful recovery

After the failure was addressed, the pipeline was run again and completed successfully. Figure 4 shows the successful run, which can be compared against the earlier failed run in Figure 2.

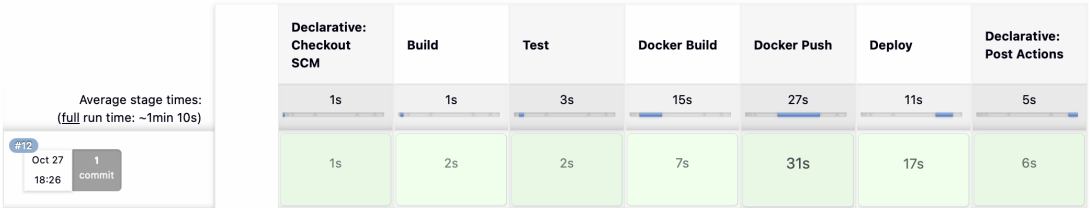


Figure 4: Pipeline run completed successfully after the failure.

4 Testing

The final test count for the project is **12**. Figure 5 shows the final test result for the project.

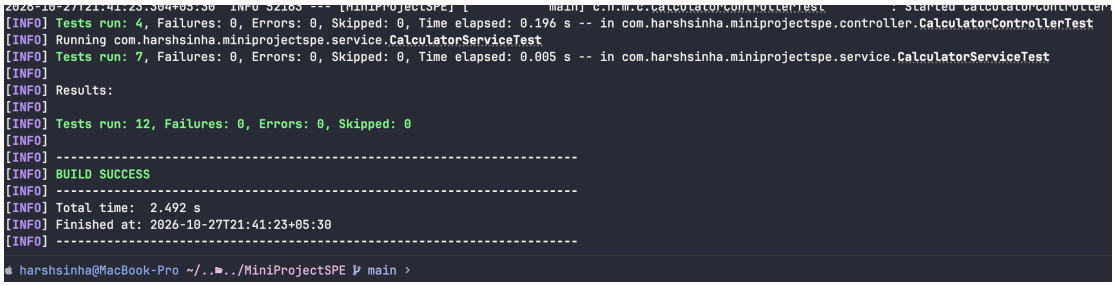


Figure 5: Final test result.

5 Repository and Image References

Item	Reference
GitHub repository	https://github.com/iHarshSinha/SPEMiniProject
Git branch	`main`
Docker Hub image	`iharshsinha/calculator-api`
Final test count	12

Table 1: Project references.

6 Conclusion

The project delivers a Spring Boot calculator REST API with a CI/CD workflow that detects a failing run, notifies by email, and confirms recovery with a successful run. The

source repository, branch, Docker Hub image, and final test count of 12 are recorded in this report.